

Fair Territory Division in Two Dimensions

The spatial quantile is a ratio quantile

Abstract

A companion to *Fair Territory Division on a Line*, extending the equal-gain and Kalai–Smorodinsky constructions from an interval to a planar territory. The equal-gain condition generalises verbatim as a statement about a mixture *measure*, but the inverse CDF that solved it in one dimension has no analogue, because there is no order on the plane. We show the correct generalisation is a quantile of the utility ratio under the mixture measure, that the fragility term splits into two distinct sensitivities, and that the monotone likelihood ratio condition is replaced by connectedness of superlevel sets. One consequence reverses a conclusion of the one-dimensional note: bimodal books, which necessarily fragment a territory on a line, do not disconnect it in the plane. Headroom is taken net throughout.

1 Setup

Let $\Omega \subset \mathbb{R}^2$ be the common territory. The primitives are as in the one-dimensional note, with position now a vector: sales densities f_a, f_b with totals S_a, S_b , opportunity density m with total M , transfer capture θ and headroom credit λ . Under the net convention,

$$c_1 = 1 - \lambda, \quad c_2 = \theta(1 - \lambda), \quad \Lambda = c_1 + c_2 = (1 - \lambda)(1 + \theta), \quad (1)$$

and the utility densities are unchanged in form:

$$u_a(s) = c_1 S_a f_a(s) + c_2 S_b f_b(s) + \lambda M m(s), \quad (2)$$

$$u_b(s) = c_2 S_a f_a(s) + c_1 S_b f_b(s) + \lambda M m(s). \quad (3)$$

Nothing in (1)–(3) refers to the dimension of the domain.

2 Equal gain is a condition on a measure

For a partition $\Omega = \Omega_a \sqcup \Omega_b$, gains against pre-merger production are

$$G_a(\Omega_a) = \int_{\Omega_a} u_a - S_a, \quad G_b(\Omega_b) = \int_{\Omega_b} u_b - S_b. \quad (4)$$

The collection step of the one-dimensional argument uses only the linear structure of (2)–(3), so it survives intact.

Proposition 1 (Equal gain fixes a measure). *Define the mixture measure Φ on Ω by*

$$d\Phi = \frac{\Lambda(S_a f_a + S_b f_b) + 2\lambda M m}{D} ds, \quad D = \Lambda(S_a + S_b) + 2\lambda M. \quad (5)$$

Then $G_a(\Omega_a) = G_b(\Omega_b)$ if and only if

$$\Phi(\Omega_a) = q, \quad q = \frac{S_a(1 + c_2) - \lambda S_b + \lambda M}{D}. \quad (6)$$

The quantile level q is unchanged from one dimension: it depends only on totals and parameters, both of which are dimension-free. What changes is that (6) is a condition on the *measure* of a region, not on a coordinate.

3 The selection problem

In one dimension Φ is a CDF on $\mathbb{R}$ and (6) is inverted directly, $w^* = \Phi^{-1}(q)$. In the plane there is no order, hence no inverse: (6) is a single scalar equation with infinitely many solutions. Any region of mixture measure q equalises the gains.

Selection therefore requires a second criterion, and the egalitarian one supplies it. Transferring an element of territory from b to a raises G_a by u_a and lowers G_b by u_b , so the exchange rate is the ratio

$$r(s) = \frac{u_a(s)}{u_b(s)}. \quad (7)$$

Maximising $\min(G_a, G_b)$ takes territory in decreasing order of r . In one dimension this was invisible: the requirement that each territory be an interval pinned the region, and under the monotonicity of Section 8 the ratio order coincided with the position order.

Remark. The two conditions answer different questions, and it is worth stating both explicitly when presenting results. Equal gain determines *how much* each wholesaler receives; the egalitarian criterion determines *which* territory.

4 Main result

Let R be the distribution function of r under the mixture measure,

$$R(t) = \Phi(\{s \in \Omega : r(s) < t\}). \quad (8)$$

Proposition 2 (The spatial quantile is a ratio quantile). *The equal-gain, egalitarian-selected region is*

$$\Omega_a = \{s \in \Omega : r(s) \geq t^*\}, \quad t^* = R^{-1}(1 - q). \quad (9)$$

This is the correct reading of $w^* = \Phi^{-1}(q)$: it was never a quantile of position, but a quantile of the utility ratio, which in one dimension happens to be ordered by position whenever the ratio is monotone. Verified numerically on Gaussian books with uniform opportunity at $S_a = 3.0$, $S_b = 1.8$, $M = 40$, $\theta = 0.4$, $\lambda = 0.3$:

q	0.533027
t^*	1.010785
$\Phi(\Omega_a)$	0.533028
G_a, G_b	5.105062, 5.105029
$ G_a - G_b $	3.28×10^{-5}

5 Fragility splits in two

In one dimension a single quantity $1/\varphi(w^*)$ governed how far the boundary moved per unit of parameter change. In the plane the threshold and the region respond separately:

$$\left. \frac{d}{dt} \Phi(\{r \geq t\}) \right|_{t^*} = -R'(t^*), \quad \left. \frac{d}{dt} \text{area}(\{r \geq t\}) \right|_{t^*}. \quad (10)$$

On the reference instance,

$R'(t^*)$	4.5337	mixture mass per unit of threshold
$ d \text{area}/dt $	5.0484	territory area per unit of threshold
range of r over Ω	[0.8591, 1.2787]	
percentile of t^*	46.7	

Which of the two matters depends on what is being defended. Mass sensitivity governs how much production changes hands; area sensitivity governs how much the map is redrawn. A territory can be stable in one and volatile in the other, and the distinction does not exist in one dimension.

6 Comparative statics

The implicit-function structure carries over, with the ratio density in the denominator:

$$\frac{dt^*}{dp} = - \frac{\partial q/\partial p + \partial R/\partial p}{R'(t^*)}. \quad (11)$$

At the reference parameters:

$dt^*/d\theta$	-0.018398
$dt^*/d\lambda$	-0.049772
dt^*/dM	-0.000267

As in one dimension, λ moves the answer substantially more than θ does.

7 Boundary geometry

The boundary is the level set $\{r = t^*\}$, which by (2)–(3) is the zero set of a fixed linear combination of the three density fields:

$$S_a f_a(s) (c_1 - t^* c_2) + S_b f_b(s) (c_2 - t^* c_1) + \lambda M m(s) (1 - t^*) = 0. \quad (12)$$

Two consequences follow immediately. If the three fields are affine in position the boundary is a straight line and t^* translates it parallel to itself. If f_a and f_b are Gaussian with common covariance and $\lambda = 0$, then $\log(f_a/f_b)$ is affine and the boundary is again a line — the linear discriminant.

Positive headroom credit breaks both, because the term $\lambda M m$ enters (12) without reference to either book. Numerically, on equal-covariance Gaussians:

straight-line R^2	
$\lambda \rightarrow 0$	0.999966
$\lambda = 0.30$	0.70651

This is the planar form of the same mechanism identified in one dimension: where both books thin out, u_a and u_b collapse toward the common term $\lambda M m$ and the ratio is dragged toward 1.

8 Regularity: connectedness replaces MLR

The one-dimensional note characterised when each fair territory is a single interval via the monotone likelihood ratio property. That condition has no planar analogue: MLR is a statement about an ordered domain, and $\mathbb{R}^2$ is not ordered. The substantive question — whether each wholesaler receives one coherent territory — becomes

$$\{s : r(s) \geq t\} \text{ is connected for the relevant } t. \quad (13)$$

This is checked numerically by counting connected components; it does not reduce to a derivative condition of the kind available on a line.

Remark (Bimodality does not disconnect in the plane). On a line, a wholesaler whose book sits in two separated concentrations necessarily receives a disconnected territory. In the plane the region can route around the gap through low-density terrain. Taking f_a as an equal mixture of two Gaussians at increasing separation, with all other primitives fixed:

separation	components	components > 1% of area	boundary R^2
0.25	1	1	0.6615
0.45	1	1	0.5583
0.65	1	1	0.1961
0.80	1	1	0.1409

The territory remains connected at every separation tested, including concentrations at opposite ends of the domain. The boundary becomes markedly non-linear — R^2 falls from 0.66 to 0.14 — but it does not break.

This reverses a conclusion of the one-dimensional note. There, the *Two Metros* case forced two segments and forcing a single cut cost 8.85% of the achievable fair gain. A material part of that penalty is an artefact of the one-dimensional projection rather than a property of the problem, and it also weakens the case for imposing contiguity through additional machinery: if fair territories are naturally connected in the plane, there may be nothing to impose.

Remark (Caveat). The domain here is the unit square with no barriers. A corridor that exists in $[0, 1]^2$ may be a mountain range, a state line, or a body of water in practice. The finding should be re-tested on the real adjacency graph before it is relied upon.

9 Kalai–Smorodinsky

The attainable maxima remain scalars, $G_a^{\max} = \int_{\Omega} u_a - S_a$ and $G_b^{\max} = \int_{\Omega} u_b - S_b$, so the mixture identity of the one-dimensional note carries over with regions in place of intervals. Writing $K_a = S_a(G_b^{\max}c_1 + G_a^{\max}c_2)$, $K_b = S_b(G_b^{\max}c_2 + G_a^{\max}c_1)$, $K_M = \lambda M(G_a^{\max} + G_b^{\max})$ and $K = K_a + K_b + K_M$, the criterion is again a ratio quantile, taken under the K -weighted mixture measure at level $q_{KS} = (G_a^{\max}G_b^{\max} + G_b^{\max}S_a)/K$.

$G_a^{\max}, G_b^{\max}$	11.6040, 12.3000
q_{KS}	0.523326
t_{KS}^*	1.013195
$G_a/G_a^{\max}, G_b/G_b^{\max}$	0.427057, 0.427053

10 Relation to optimal transport

The planar two-player problem is an instance of the framework in Warren¹, whose Example 2.1 is exactly this setting: two emperors with capital cities at points x_1, x_2 in a region $\Omega \subset \mathbb{R}^2$, negotiating how to split it, each valuing a unit of land at $e^{-|x_i-y|^2/2}$, with allocations being measures summing to the region. Two observations.

The exponential form is not a restriction. Warren’s utility density is $s(x, y) = e^{-c(x, y)}$ and utility is $\int s(p_i, y) d\nu_i(y)$ — linear in the allocation measure, as here. Since $u_a > 0$ throughout Ω (the term λMm guarantees it), setting $c(a, s) = -\log u_a(s)$ recovers (2) exactly. It is a reparameterisation.

The disagreement point is a genuine gap. Warren’s construction takes the disagreement point to be the zero allocation, which is what produces the clean identification of Nash bargaining solutions with optimal transport plans and the entropy-plus-transport functional. The baseline used here is pre-merger production. The first-order condition still yields a ratio threshold, but that functional is not the right object, so the transport machinery does not come along.

For two players, Proposition 2 is exact and computationally trivial, and optimal transport adds nothing. Its value is for $N > 2$ wholesalers with a geographic cost, where semi-discrete transport with capacity constraints produces Laguerre cells and hence contiguity by construction. Section 8 suggests that may be solving a problem the plane does not have.

11 Scope and what to check

Covered. Two wholesalers, planar continuum territory, equal-gain and Kalai–Smorodinsky selection against pre-merger production, arbitrary sales and opportunity densities, net headroom.

Not covered. Capacity, travel and balance constraints; three or more wholesalers, where no threshold rule exists; indivisibility of postal codes, for which the continuum result is a relaxation and the residual scales with book concentration.

Three checks on real geography.

1. **Do superlevel sets stay connected?** Compute r per postal code, threshold at t^* , and count connected components on the true adjacency graph. This is the planar regularity condition and it decides whether contiguity needs to be imposed at all.
2. **Which fragility dominates?** Report $R'(t^*)$ and the area derivative side by side. They can diverge, and the one that matters depends on whether production or the map is being defended.
3. **How non-linear is the boundary?** A low straight-line R^2 is not a failure — it is the signature of headroom credit doing work — but it predicts how hard the result will be to explain from a map.

¹M. Warren, *Continuum Nash bargaining solutions*, Nonlinear Differ. Equ. Appl. **32**:109 (2025), open access. doi:10.1007/s00030-025-01118-7.

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